LG TV Conversion - Coding Session Plan

Steps 4-5 only | Team work session before Tutorial I | 4 November 2026

Today's objective

Make the LG-Samsung TV comparison defensible before moving to the Logit. By the end of the session, the team should have a reproducible cohort audit, verified KPI numerators/denominators, evidence on price overlap and concentration, and clearly labelled sensitivity samples.

Scope for today

The interim slides are already complete. The step-by-step plan places Steps 4-5 in the 1-5 November window and asks the group to document exactly what enters the analysis, verify journey KPIs, assess price overlap and concentration, and prepare the evidence to discuss at Tutorial I. The model sequence itself starts in Steps 6-7, so today should prioritise measurement and comparability rather than adding extra techniques.

Reference values to reproduce - not numbers to hard-code

Check	LG	Samsung	Total / note
Eligible TV consideration rows	22,213	52,499	74,712
Purchase outcomes	505	1,645	2,150
View-to-purchase	2.27%	3.13%	Recompute from counts
Distinct TV products	105	91	196

End-of-session deliverables

Done	Task	Required output
■	One clean analytical table with a unique user_id x session_id x product_id key.	Saved or reproducibly generated from raw events.
■	Cohort audit from raw events to final TV LG/Samsung rows.	Counts, exclusions and data-quality notes.
■	Verified KPI table with numerator, denominator and rate.	Overall + monthly; include view-to-cart and cart-to-purchase.
■	Price-overlap evidence and concentration diagnostics.	Plot + quantiles + product/user/month summaries.
■	Clearly labelled sensitivity samples.	Full cohort preserved; restricted samples never silently replace it.
■	Short list of open methodological questions.	Ready for Tutorial I on 5 November.

Do not spend today on: final recommendations, extra models, segmentation/PCA, or polishing final-report prose.

Step 4 - Audit the analytical cohort

Goal: another team member should be able to rebuild the final table from the raw CSV using the written rules.

4.1 Build an explicit audit trail

- **Load the raw behavioural file unchanged.** Keep paths relative and document the expected input file.
- **Parse event_time** and check the full date range.
- **Audit missing values, invalid/non-positive prices, exact duplicate event rows, and brand/category consistency by product_id.** Do not silently drop anything.
- **Filter to TVs, then LG/Samsung, then construct the eligible cohort.** Record the row/event count after every stage.
- **Analytical key:** user_id x session_id x product_id. Require at least one view. Use the first view as the starting point and retain its price and time.
- **Outcome Y:** 1 only if the same product is purchased strictly after the first view within the same analytical key; otherwise 0.
- Keep user_id in the key because the previous audit found four session IDs shared across users.
- Preserve and report the previously identified 1,247 duplicate event rows. Verify whether deduplication changes any first-view/presence-based result before deciding how to handle them.

4.2 Required validation checks

Done	Task	Required output
■	Assert one row per user-session-product in the analytical table.	No duplicated analytical keys.
■	Assert Y is binary and contains no missing values.	Only 0/1 outcomes.
■	Count purchases without an eligible prior view.	Documented separately, not forced into the cohort.
■	Count equal-timestamp view/purchase or cart/purchase cases.	Flag for later timing sensitivity; do not improvise a rule silently.
■	Check product_id maps consistently to brand/category.	Any inconsistency documented before modelling.

Interpretation rule

A row with Y=0 means only that no qualifying purchase was observed after the first view within that analytical unit. It does not mean the shopper never purchased elsewhere, in another session or after the data window.

Suggested notebook output

Create an **Audit Summary** table with columns such as: stage, number of raw events, number of analytical keys, number of users, number of products, purchases retained, exclusions and notes. Follow it with a short Markdown cell stating the final cohort definition in plain English.

Step 4 - Rebuild and verify the journey KPIs

The purpose is not to re-copy the interim percentages. Recompute every rate from explicit numerators and denominators.

KPI definitions to encode beside the code

KPI	Numerator	Denominator	Reference
View-to-purchase	Eligible rows with purchase strictly after first view	All eligible viewed rows	LG 2.27% | Samsung 3.13%
View-to-cart	Eligible rows with cart strictly after first view	All eligible viewed rows	LG 4.83% | Samsung 6.31%
Cart-to-purchase	Rows with purchase after an eligible cart	Rows with an eligible cart	LG 34.86% | Samsung 40.17%

What to check

- For every KPI, print **numerator, denominator and percentage**. Never store only the final percentage.
- Compute the same KPI table overall and by month (October and November 2019).
- Count purchases with no qualifying cart separately. The intermediate rates do not have to multiply to overall view-to-purchase conversion.
- Manually inspect several event sequences for each pattern: view -> purchase; view -> cart -> purchase; repeated views; cart without purchase; purchase without qualifying cart; equal timestamps.
- If your numbers differ from the interim slides or reference audit, investigate filters, keys, timestamp ordering and denominator definitions before changing the result.

Minimum KPI output for today

Done	Task	Required output
■	Overall KPI table by brand.	Counts + rates for all three journey measures.
■	Monthly KPI table by brand.	October and November counts + rates.
■	Manual sequence audit.	A few concrete rows showing that code matches definitions.
■	Short discrepancy note, if needed.	Explain any mismatch rather than overwriting it.

Important

The final modelling outcome is view-to-purchase. View-to-cart and cart-to-purchase are diagnostic KPIs that help locate where further investigation may be useful; they are not separate causal explanations.

Step 5 - Check comparability and concentration

Goal: determine whether the LG-Samsung comparison depends on non-overlapping price ranges, a few dominant products, highly active users or one month.

5.1 Price overlap

- Use **initial price at first view** from the analytical table, not every repeated event price.
- Plot the full LG and Samsung price distributions on the same scale.
- Report at least a small quantile table. A useful choice is min, 5th, 25th, 50th, 75th, 95th percentile and max.
- Identify the range where both brands have meaningful support. Do not define the range only because it preserves a desired gap.
- Remember: price overlap does not make TVs equivalent in screen size, technology, quality or availability; those characteristics are unobserved.

5.2 Product concentration

- For each product_id and brand, compute number of consideration rows, number of purchase outcomes and conversion rate.
- Calculate the share of brand observations explained by the top 1, top 5 and top 10 products.
- Reproduce the prior diagnostic that the top five products account for about **42.3% of LG rows and 52.4% of Samsung rows**.
- Keep separate rankings for **most viewed** and **best selling** products. They are not the same concept.

5.3 User and month concentration

- For each user, count analytical rows, sessions, brands viewed and purchase outcomes. Report the distribution and the share contributed by the most active users.
- Document how many users contribute multiple analytical rows. This will matter later for inference/clustered uncertainty.
- By month, report observations, purchases, products, users and KPI rates. The interim already shows the gap has the same sign in October and November; now verify the underlying counts.

Outputs required before stopping

Done	Task	Required output
■	Price-overlap plot + quantile table.	Clear common-support discussion.
■	Product concentration table.	Top product shares, purchases and conversion.
■	User concentration summary.	Repeated-user contribution documented.
■	Monthly composition table.	Counts and KPI rates by brand/month.

Step 5 - Prepare sensitivity samples and Tutorial I

Sensitivity work today should be simple and transparent. Preserve the full cohort as the baseline and create restricted samples only to answer a stated concern.

5.4 Sensitivity samples to prepare

Concern	Sample/check	What to report
Price comparability	Restrict to a justified common-price range once overlap is inspected.	Exact rule, N, purchases, KPI changes and narrower population.
Dominant products	Create a sample excluding each brand's most-viewed product; optionally prepare top-product variants for later.	Removed products, N, purchases and change in raw KPI gap.
Repeated users	Document repeated-user structure now; do not invent the final inference rule before tutor feedback.	How many users repeat, how much of the sample they generate, questions for tutor.
Timing ambiguity	Flag equal timestamps and end-of-window exposure cases. Full sensitivity can follow after tutor feedback.	Counts of ambiguous/boundary cases and proposed alternative rule.

Rule for restricted samples

- Name each restricted sample explicitly, e.g. **baseline_full**, **common_price**, **exclude_top_viewed**.
- After any restriction, recompute denominators, purchases and KPIs from scratch.
- Never overwrite the baseline table or keep using the old denominator.
- A restricted sample answers a narrower question. Record that change in interpretation.

What to bring to Tutorial I - 5 November

Done	Task	Required output
■	Primary RQ and exact cohort definition.	One short statement everyone can repeat.
■	Reproduced summary and KPI table.	Numbers verified from team notebook.
■	Price-overlap evidence.	Plot + quantiles + interpretation.
■	Concentration evidence.	Products/users/months; main comparability concern.
■	Proposed M1/M2 model sequence.	Brand only, then brand + log(initial price) + month; modelling starts after these checks.
■	Three targeted questions.	Unit of analysis; comparability/control choice; repeated-user/product dependence.

Stop condition for today

The team is ready to stop when it can explain what one row means, reconstruct every KPI from its denominator, describe where LG and Samsung are or are not comparable, and identify the methodological choices that genuinely require tutor feedback.

Basis: LG_Project_StepbyStep.pdf (Steps 4-5 and Tutorial I), cross-checked against the submitted interim slides. Coding-session work plan; not final-report text.